



Science and Technology

QUOTES

Technology is a useful servant but a dangerous master
– Nobel Prize Winner Christian Lous Lange

GENERAL CONCEPTS

General definitions of Science and Technology
Mankind's uses of Science and Technology
The need and dangers of scientific research and advancement

OVERARCHING THEMES

Theme 1: Science VS Ethics

Theme 2: Regulation of Scientific advancement

Theme 3: Artificial Intelligence – to be feared or welcome?

LEARNING OBJECTIVE

This is the info pack under the topic of **Science and Technology**, where the concepts of **Scientific Research, Artificial intelligence, Technology** and its applications to man

Important concrete topic! A MUST to study.



3 main themes



Quotes are easy to use



Examples are easy to remember and interchangeable



Question types and overarching themes are easy to spot to spot

PAST YEAR QUESTIONS

To what extent can the regulation of scientific and technological developments be justified? (2014)

Does modern technology always improve the quality of people's lives? (2006)

'Science is unreliable, being based as much on theory as on fact.' Is this a fair comment? (2011)



GENERAL CONCEPTS

- *Technology is a useful servant but a dangerous master.* – Nobel Prize Winner Christian Lous Lange
 - When it comes to learning to live in a world dominated by technology, we are opening ourselves up to a world of opportunity, connection, experience and learning at a pace and scale unprecedented.
 - At the same time, we are exposing ourselves to a world potentially dominated by overuse, disconnect, fraud, vile and inappropriate content, predators, haters and a depreciation of self-worth.

DEFINITIONS

- Science refers to the **systematic study of the structure and behaviour of the physical and natural world** through observation and equipment. Historically, the progress of science has been driven by the pure curiosity to understand the world.
- Technology refers to the application of the scientific knowledge for practical purposes.
- Aims of both are mainly to
 - Acquire knowledge and improve understanding of the natural and physical world
 - Solve problems or improve pre-existing solutions to problems
 - Make life more convenient and comfortable
 - Improve the quality of life

Theme 1: SCIENCE VS ETHICS

Assisted Reproductive Technology (ART)

- **ART** is the technology used to achieve pregnancy through procedures such as fertility medication, artificial insemination, in vitro fertilization (IVF) and surrogacy. Such technology has enabled people who are unable to have children naturally to have children of their own.
- IVF
 - Issues related to IVF include the possibility of a mix-up of sperm and eggs, which has happened in a number of countries.
 - In Singapore, Thomson Medical's fertility centre made a blunder in its IVF procedures in 2010 and fertilised the woman's extracted eggs with another man's sperm, instead of her husband's. This was only discovered after the woman gave birth to a baby girl.
 - There is some controversy over much older women seeking IVF treatment, with concerns about the health risks and the welfare of the child as he grows up.



- This has prompted Quebec (a province in Canada) to consider a ban on IVF for women above the age of 42.

- **Surrogacy**

- Surrogacy is when another woman carries and gives birth to a baby for the couple who want to have a child.
- Commercial surrogacy is legal in countries such as India and Thailand, and in California in the US. While it is not legal in China, illegal agencies offer surrogacy. Issues of exploitation of poor women abound in commercial surrogacy. Critics say the intended parents are exploiting poor women by hiring them at a cut-rate cost to undergo the hardship, pain and risks of labour. The country is also facing an increasingly booming underground market in surrogate motherhood.
- In late 2016, a human trafficking crackdown brought to light a syndicate hiring Filipino women to travel to Cambodia to act as surrogate mothers for foreigners.
- Case of surrogacy gone wrong
 - In 2014, an Australian couple was accused of abandoning their baby with his Thai surrogate mother after discovering the child had Down syndrome, and taking home his healthy twin. This case led to Thailand banning commercial surrogacy for foreigners in 2015, as well as a ban on the use of agents, or any promotion of women willing to be surrogate mothers.

Biotechnology

- Biotechnology, broadly defined, is the use and modification of living organisms for various purposes.
- The modern usage of biotechnology often includes genetic engineering and cell and tissue culture technologies to
 - combat debilitating and rare diseases, particularly by tackling the root (genetic cause) of these diseases
 - reduce our environmental footprint with the use of less and cleaner energy, and safer, cleaner and more efficient industrial manufacturing processes,
 - deal with world hunger by creating foods with better nutritional value and that last longer, developing disease and pest-resistant crops,
 - slow down the ageing process.

GM food

- **Developments & controversies**
 - Bt-corn is a form of sweet corn that has been genetically modified to include an insect-killing gene. This means that farmers do not have to spray with pesticides, because the insects die from eating the corn. No spraying means less harm to the environment and the workers



handling the toxic spray. The move has caused debate, however, because the same gene that attacks corn predators also appears to kill the Monarch butterfly.

- Golden rice was first created to fight vitamin A deficiency, which affects 250 million people around the world and can cause blindness and even death.

- **Regulation**

- The official regulatory agency for GM foods is the Agri-Food and Veterinary Authority (AVA). GM foods commercially available in Singapore have undergone various lines of scrutiny for safety. Laboratories have been set up by the AVA to test such food items and make sure they do not pose any health risks for people here. About half of the soya and a third of the corn sold here is genetically modified, but they have all been tested stringently before hitting the shelves. There are currently no legislations and guidelines for the labelling of GM foods in Singapore, which is not mandatory.

Reproductive cloning

- **Reproductive cloning** is the production of a genetic duplicate of an existing organism. Currently, reproductive cloning generally uses somatic cell nuclear transfer to create animals that are genetically identical.
- **Animal reproductive cloning**
 - First success was Dolly, the lamb cloned from the udder cell of a 6-year-old sheep.
 - Many applications of animal cloning and genetic modification are controversial for environmental, health, animal welfare, and social reasons.
 - Reproductive cloning is a very inefficient technique and most cloned animal embryos cannot develop into healthy individuals. For instance, Dolly was the only clone to be born live out of a total of 277 cloned embryos. This very low efficiency, combined with safety concerns, presents a serious obstacle to the application of reproductive cloning.
 - Researchers have observed some adverse health effects in sheep and other mammals that have been cloned. These include an increase in birth size and a variety of defects in vital organs, such as the liver, brain and heart. Other consequences include premature aging and problems with the immune system.
 - Nonetheless, animal cloning efforts continue. Many of these efforts are part of scientific experiments to better human life. Reproductive cloning may enable researchers to make copies of animals with the potential benefits for the fields of medicine and agriculture.

Euthanasia

- Euthanasia is the termination of a person's life in order to relieve him of his suffering.



- As of December 2017, Victoria has now joined the ranks of the Netherlands, Canada, Belgium, Colombia and Luxembourg which have legalised euthanasia. Several states in the United States, including California, Washington, Vermont and Oregon, have passed laws allowing assisted suicide.
- In physician-assisted suicide, a physician supplies a lethal drug or suggests a means by which a patient will use it to complete the act that brings about death.
- In euthanasia, the physician performs the act of directly injecting the medication to cause death. These terms do not apply to a patient's refusal of life-support measures, or to the request for these life-prolonging measures to be withdrawn.
- One of the biggest - and perennial - concerns of legalising physician-assisted suicide and euthanasia is that it will lead to the "**slippery slope**". The slippery slope argument contends that even a relatively conservative law like the Victorian model would evolve over time, becoming more permissive and with the right to receive assisted suicide extending to many others: from physician-assisted suicide for the terminally ill and fully consenting adults to euthanasia for patients who cannot give consent - the unconscious, severely disabled children, and the mentally ill.
 - Example: Belgium, which legalised euthanasia in 2002, further amended its law in 2014 to allow for the euthanasia of terminally ill children. The law also allows euthanasia for patients suffering "unbearably" from any "untreatable" medical condition, terminal or non-terminal, including mental disorders. The law has been used for patients with autism spectrum disorder, eating disorders, bipolar disorder, and major depression, which many doctors find disturbing.
- The slippery slope argument also raises a **practical concern**: If assisted suicide were legalised, there must be regulations to ensure that a patient's decision for suicide is informed, competent, and made without any external coercion. There must be a tight process of robust vetting, assessment and consultation. The doctor has to provide detailed information, including the diagnosis and prognosis, and possible treatment options, and assess the patient's mental capacity to make that fateful decision.
- But the danger is that such regulations and safeguards cannot be adequately enforced in practice, and that particularly vulnerable patients - for example, the aged and those with severe disabilities - might be pressured into accepting assistance in dying when they do not really want it.
- There may be that additional indirect pressure when the onus of responsibility is being shifted to these patients who may then be seen to have the power to end their suffering; and refusing would be seen as the patient's own doing - something which might even engender resentment in overwrought caregivers who are physically, emotionally, and financially exhausted.

ROBOTICS

- Robotics is a branch of engineering that involves the conception, design, manufacture and operations of robots.



- The development and commercialisation of robotic helpers will greatly enhance the way we complete tasks. This runs the whole gamut from robo-nannies taking care of our babies while we are at work to Robochef which mimics the movements of eminent, top-tier chefs to prepare exquisite dishes in the comfort of our homes. As it is, there are already small lightweight robot vacuum cleaners and floor-washing robots used at home. Toyota and Honda expect domestic robots to become a huge market in future, with robots working as domestic helpers and are investing in developing such robots. Such machines can replace menial labour by performing tedious jobs that few are willing to undertake and alleviate problems associated with an ageing population prevalent in many developed countries.

ARTIFICIAL INTELLIGENCE

- The simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning and self-correction.
- With the rapid advancements in artificial intelligence, human beings now enjoy greater convenience by using machine learning - computer systems that can learn from data (machine learning), instead of blindly following instructions. Technology looks set to take on increasingly personal roles in people's lives by learning human habits and predicting people's needs.
 - (1) Apple's digital assistant Siri which can answer simple questions, conduct web searches and other basic functions
 - (2) Driverless cars
 - (3) Google Now can show traffic information during your daily commute and give you a shopping list reminder when you are at the store.
- Stephen Hawking and Elon Musk have both warned that AI, once it surpasses human intelligence, could pose a threat to the existence of human civilisation. With the rise in processing power and the interest in scientists to develop "deep learning systems" which enhance robots and machines' ability by imitating the layers of neurons in a human brain and processing vast amounts of data at high speed, robots and machines may be better able to perform all tasks than human beings in future. There are concerns about "the technological singularity" when intelligent machines take off in a rapid cycle of self-improvement, leaving mere humans behind.
- The Orwellian dystopia of a "Big Brother" constantly watching Mankind now seems chillingly realistic.
 - The development of commercial drones and miniscule cameras has increased the ease of spying.
 - The confluence of such technology and the fact that drone regulations and anti-drone technology is still in its infancy poses a great threat to both our current and future definitions of privacy.



- The Edward Snowden incident revealed a shocking compromise between national security and individual privacy that has yet to be resolved, partly due to the dynamic nature of technological change which constantly surfaces new moral and legal issues.
- This issue could be further exacerbated by the development of robotics and artificial intelligence challenging on moral and social grounds involving our notions of privacy and its centrality to security issues.

SINGAPORE

Current trends

- These are some efforts that will be given a boost in Singapore's science and technology plan, which will see public spending on research, innovation and enterprise increase to S\$19 billion for 2016 to 2020 — 18 per cent more than the S\$16.1 billion spent from 2011 to 2015.
- At the announcement of the Research, Innovation and Enterprise 2020 (RIE2020) plan, PM Lee stressed the importance of research and innovation in growing the economy, helping workers to thrive amidst technological changes and in making healthcare and the living environment better for Singaporeans.
- Four domains have been identified in RIE2020 based on areas of competitive advantage or national needs. They are advanced manufacturing and engineering, health and biomedical sciences, services and digital economy, and urban solutions and sustainability.
- Funds will also be used to boost academic research, sustain a strong base of science and engineering talent and grow innovative enterprises.
- Government helping SMEs
 - For example, local startup DC Frontiers was awarded a contract for its Handshakes platform, an AI-powered tool that analyzes networks and connections between people and companies to identify things like legal ownership, conflicts of interest, and potential fraud.
- New technology
 - National Library, which will use various technologies like wall projections to enhance narratives. A virtual library containing ebooks, audiobooks, and virtual exhibitions is also in the works.
 - MyMaritime@SG, in the meantime, spotlights Singapore's port and marine specialization. The system will make it easier to request marine services, permits, and approvals. The



Maritime Single Window platform will be used for submitting port-related documentation for port clearance.

Smart nation

- In 2015, the government announced the 'Smart Nation Initiative' with an aim to see Singapore become the world's first smart nation by 2025, and also further enhance networked computerisation capabilities in order to remain globally competitive.
- In October 2014, the Singapore government announced that around 1,000 sensors would be installed throughout the island to track anything from air quality and water levels to public safety.
- For example, the recently announced Singapore Student Learning Space (SLS) for various primary and secondary schools is an initiative that empowers students towards self-directed learning.

National security

- For instance, an Automated Biometric and Behavioural Screening Suite (ABBSS) has been deployed at Woodlands Checkpoint, a land border crossing between Singapore and Malaysia. ABBSS encompasses a state-of-the-art video analytics system, which can perform real-time facial recognition, cross-referencing that against a large database.



Theme 2: REGULATIONS OF SCIENTIFIC DEVELOPMENTS

- Regulation is on grounds of safety, environment and ethics
- ****BALANCE**** : Governments should regulate scientific research so that the research is beneficial to the masses, but governments should not control to the extent of hindering the progress of the research.

SCIENCE - CONTEMPORARY DEBATES/PRESSING ISSUES

Sub-Issue 1: Increased scepticism in science due to its faults

Background:

- Scientists make false claims on their reports especially in areas like biomedical science in order to get published or earn a higher income. There are inadequate safeguarding checks.
- A study published in the proceedings of the National Academy of Science found that two-thirds of the 2047 papers that were published had to be retracted due to scientific misconduct.
- Rats used by many scientists nowadays are overfed, understimulated and in some cases pumped through with antibiotics are placing unseen constraints on scientific research. These animals respond differently from humans, yet the same drugs are still fed to us.

Arguments/Viewpoints/Perspectives

- Those who disapprove of science use this as a case of how many scientists now pursue science not for knowledge but other reasons such as for fame and money. In health care, research is often conducted at the behest of companies that have a large financial stake in the results. Scientists are willing to publish reports as quickly as possible and hence compromising on the quality of work.
- Those who still believe in science argue that what science really needs is greater enthusiasm for those people who are willing to invest the time to try to sort the truth from hype and bring that to the public. Academic science does far too little to encourage such voices. At the same time, it is facile to dismiss science itself. The most careful scientists, and the best science journalists, realize that all science is provisional. There will always be things that we haven't figured out yet, and even some that we get wrong. But science is not just about conclusions, which are occasionally incorrect. It's about a methodology for investigation, which includes, at its core, a relentless drive towards questioning that which came before.



Application to SG:

- Singapore has established bodies like the Health Sciences Authority for monitoring clinical drug trials and the Biomedical Research council.
- Scientists in SG have developed a drug-delivery system using epigallocatechin gallate (EGCG), a key ingredient in green tea, that can kill cancer cells more effectively than traditional cancer treatments. The hydrogel was tested on tumour-bearing mice, and tumours shrank 77 per cent over 28 days.

Sub-Issue 2: Research Ethics

1. Case Study: Recreating avian and Spanish flu in labs

Quick Background:

- Scientists have created a life-threatening virus that closely resembles the 1918 Spanish flu strain that killed an estimated 50m people in an experiment.

Arguments:

- US researchers said the experiments were crucial for understanding the public health risk posed by viruses currently circulating in wild birds. Proponents believe this could give us an early warning system for the next pandemic, and could aid in the timely creation of a vaccine.
- Critics condemned the studies as dangerous and called on funders to stop the work. Scientists should not take such risks without strong evidence that the work could save lives, which this paper does not provide
- There were also concerns that blocking the study would pave the way for government restriction of academic freedom.

Application to SG

- A Singaporean worker was infected with SARS in 2003 whilst researching on it.
- The Agency for Science, Technology and Research (ASTAR) follow a strict set of rules of research ethics to ensure that in the pursuit of scientific progress, huge risks are not taken in order to protect the wellbeing of the people.

2. Case Study: Hwang Woo-Suk Cloning scandal

Quick Background:



- Professor Hwang Woo-Suk, a highly regarded, highly funded South Korean researcher at Seoul National University, achieved international fame for his work on embryonic stem cells and the promises his findings offered
- His reputation quickly unraveled, however, and his research activities were halted when his success in somatic cell nuclear transfer (SCNT) became mired in scandal, particularly when it emerged that many of his data on SCNT were fabricated.

Stem Cell debate:

- People in support of stem cell research feel that it is filled with great potential and creativity. They believe many of the claims that stem cell researchers make because they want them to be true and something repeated so often eventually comes to sound plausible. The scandal has led to the search for alternative methods of deriving tailor-made, pluripotent stem cells which can potentially prove to be extremely beneficial to mankind due to the ability of such cells to develop into any other cell. Following the scandal, it has led to more public scrutiny of future research findings.
- People who feel that stem cell research ought to stop think that the stem cell debate is simply saturated with falsehoods, exaggerations, and unfounded claims.

Sub-Issue 3: Justifiability of scientific research

1. Case study: Higgs Boson

Quick Background:

- The Higgs boson or Higgs particle is an elementary particle in the Standard Model of Particle physics. Its main relevance is that it is the smallest possible excitation of the Higgs field – a field that unlike the more familiar electromagnetic field cannot be "turned off", but instead takes a constant value almost everywhere.
- The existence of the Higgs boson has yet to be confirmed

Arguments:

- Stephen Hawking in the preface of his book *Starmus* wrote, "The Higgs potential has the worrisome feature that it might become metastable at energies above 100bn gigaelectronvolts. This could mean that the universe could undergo catastrophic vacuum decay, with a bubble of the true vacuum expanding at the speed of light. This could happen at any time and we wouldn't see it coming."



- People who are apathetic towards the Higgs boson feel this way because it is not going to be used to cure diseases or improve technology. It is also not worth the billions of dollars that it costs to build particle accelerators.
- Physicists feel that such research is extremely important as it pushes modern technology to its limits, often yielding new technology of great practical importance. For example the World Wide Web was developed after elementary particle physicists developed a method of sharing data. Although the Higgs boson is pure science, it has the potential to lead to great discoveries and spinoffs.

Sub-Issue 4: Profits and funding in science

- Science projects and research has brought about a great deal of prestige to a country. Hence countries tend to invest heavily in scientific research.
- However developing countries, due to a lack of funds and science not being a priority, devote a significantly smaller proportion of their budgets to scientific pursuits.
- According to OECD, around two-thirds of research and development in scientific and technical fields is carried out by industries, and 20% and 10% respectively by universities and government. Comparatively, in countries with less GDP, such as Portugal and Mexico the industry contribution is significantly lower.

Case study: Science and the EU

- The EU's focus used to be in training scientists and helping them move seamlessly throughout the region. However the EU is now investing more in research. While only 30% of the proposals put forth receive money, countries that put money into EU research investment actually gain. Between 2007 and 2013, roughly 12800 participants from the UK have received over 4.9 billion Euro in funding.

Arguments

- Economic or other self-interests (driven by the profit motive) may blind some people to continue research into unsafe or unethical practices even though they may be cognisant about the negative impacts. In 2010, the WHO warned the endemic problem of corruption in the pharmaceutical industry. Clinical trials were conducted without proper regulatory approval. Drugs were produced through substandard or counterfeit methods, leading to products that are less effective at best and hazardous at worst. This lead to patients suffering and have direct life or death consequences. This illustrates how unscrupulous researchers and corrupt businesses do not care about where to



draw the line as they are primarily enticed by the monetary benefits and disregard the welfare of individuals.

- Such investment into scientific research is necessary to provide jobs for many budding scientists and also to ensure the progress and development of science. Without such funding, there could be little or no progress in science. Obama has pointed out that research and innovation could provide many jobs for the unemployed.

Sub-Issue 5: Public purpose of science

Background:

- During the war, the Office of Scientific Research and Development (OSRD) had virtually unlimited resources and the authority to hand out contracts as it saw fit. Their aggressive investments in new technologies quickly paid off. Penicillin and improved surgical techniques saved the lives of countless soldiers fighting abroad.
- Before the war, scientific research had primarily relied on private foundations and wealthy benefactors. The OSRD was incredibly effective, but its undemocratic organization and operations were ill-suited to peace.
- According to Vannevar Bush, the director of OSRD, the answer was a resounding yes. At the behest of Pres. Roosevelt, he prepared a report, "Science, the Endless Frontier," in which he argued that America's relatively prosperous and militarily secure future required a federal program to train and support a large number of scientists and engineers.

Arguments:

- Scientists argue that all types of research are important. It's a pipeline which, when effective, acts as a well-spring of ideas that broaden knowledge that, occasionally, sprouts incredibly useful applications that generate wealth that helps support more investment in discovery.
- Critics argue that this process has been examined to death but by its very nature is unpredictable. All we can do is ensure that the best quality is achieved at every level and that we do not inadvertently limit the resources and ideas of those who ultimately practice this bizarre but essential driver of progress. There is no certainty in the research of science.



Application to SG:

- In Singapore, weaponry research and the likes are vital to our survival as a small nation and should be justified to a certain extent. Ideally, there is not a need to venture into nuclear and chemical weaponry since we are a country that is always on the defensive stance where we are prepared in cases of war and conflict. As such, the government does favour medical research which keeps our citizens safe and is sufficient to the success of our nation. For example, scientists from the Massachusetts Institute of Technology (MIT) and the Singapore-MIT Alliance for Research and Technology (Smart) have contributed to a new breakthrough in dengue research that could help patients recover more quickly and protect them from developing the more severe form of the infection by developing a new antibody in 2013.